open-bpm Benchmark Report

GiantSteps Tempo Dataset Evaluation

Project	open-bpm v0.1.0
Algorithm	Triple-estimator fusion (IOI histogram + comb filter + autocorrelation)
Dataset	GiantSteps Tempo Dataset (664 tracks, electronic music)
Date	2026-04-09
Platform	macOS (Apple Silicon), Rust release build

---

1. Summary

Metric	Value
Tracks tested	664
Missing audio	0
Detection errors	0
Acc1 (4% tolerance)	**457 / 664 (68.8%)**
Acc2 (octave-tolerant)	**523 / 664 (78.7%)**
Octave errors	66 (9.9%)

Acc1 measures the percentage of tracks where the detected BPM falls within 4% of the ground-truth tempo. **Acc2** extends this tolerance to include octave multiples (2x, 0.5x, 3x, 1/3x), which is standard in MIR tempo evaluation.

---

2. Error Distribution

2.1 Absolute Error Statistics

Statistic	Value
Mean absolute error	16.36%
Median absolute error	0.36%
90th percentile	52.11%
95th percentile	100.00%
99th percentile	106.67%
Maximum error	213.26%

2.2 Accuracy at Varying Tolerances

Tolerance	Correct	Accuracy
<= 1%	407 / 664	61.3%
<= 2%	436 / 664	65.7%
<= 4% (Acc1)	457 / 664	68.8%
<= 4% octave-tolerant (Acc2)	523 / 664	78.7%

The tight cluster of 61.3% of tracks within 1% error indicates high precision on correctly-identified tracks. The gap between Acc1 and Acc2 (10 percentage points) reveals that octave confusion is a significant error mode.

---

3. Accuracy by BPM Range

3.1 Acc1 (Strict 4% Tolerance)

BPM Range	Tracks	Correct	Accuracy
< 90	68	3	4.4%
90 – 119	41	11	26.8%
120 – 139	191	133	69.6%
140 – 159	200	176	88.0%
160 – 179	149	127	85.2%
>= 180	15	7	46.7%

3.2 Acc2 (Octave-Tolerant)

BPM Range	Tracks	Correct	Accuracy
< 90	68	54	79.4%
90 – 119	41	14	34.1%
120 – 139	191	141	73.8%
140 – 159	200	180	90.0%
160 – 179	149	127	85.2%
>= 180	15	7	46.7%

3.3 Analysis

- **Sweet spot (120–179 BPM):** The detector achieves 69.6%–88.0% Acc1 in the most common electronic music tempo range. This is where the majority of the dataset concentrates (540 of 664 tracks, 81.3%).
- **Slow tempos (< 90 BPM):** Acc1 drops to 4.4%, but Acc2 recovers to 79.4%, confirming that most errors in this range are octave doublings (the detector returns 2x the true BPM). This is a known challenge for tempo estimation algorithms.
- **90–119 BPM range:** Both Acc1 (26.8%) and Acc2 (34.1%) are notably low, suggesting difficulty in this transitional range.
- **Fast tempos (>= 180 BPM):** Limited sample size (15 tracks) makes conclusions tentative, but accuracy is moderate at 46.7%.

---

4. Dataset Distribution

BPM Range	Tracks	Share
< 90	68	10.2%
90 – 119	41	6.2%
120 – 139	191	28.8%
140 – 159	200	30.1%
160 – 179	149	22.4%
>= 180	15	2.3%

The GiantSteps dataset is predominantly composed of electronic dance music, with 81.3% of tracks in the 120–179 BPM range typical of house, techno, trance, and drum & bass genres.

---

5. Worst-Case Errors

The 10 tracks with the largest absolute error:

Track ID	Ground Truth	Detected	Error
3980001.LOFI	53 BPM	166.03	+213.3%
1874244.LOFI	70 BPM	186.56	+166.5%
4609944.LOFI	58 BPM	144.95	+149.9%
3789981.LOFI	80 BPM	173.12	+116.4%
1479462.LOFI	90 BPM	193.96	+115.5%
4960424.LOFI	85 BPM	177.23	+108.5%
4532060.LOFI	85 BPM	175.72	+106.7%
4153394.LOFI	80 BPM	165.34	+106.7%
2704868.LOFI	64 BPM	132.18	+106.5%
3167057.LOFI	85 BPM	175.33	+106.3%

All worst-case errors occur on slow tracks (53–90 BPM) where the detector latches onto a harmonic multiple of the true tempo. The ~100% and ~200% errors correspond to 2x and 3x octave confusion respectively.

---

6. Methodology

6.1 Dataset

GiantSteps Tempo Dataset – a widely-used MIR benchmark comprising 664 electronic music tracks with manually annotated tempo values. Audio files are MP3 format.

6.2 Metrics

- **Acc1** (Accuracy 1): The detected BPM is within 4% of the ground truth.
  - Formula: `|detected - truth| / truth <= 0.04`
- **Acc2** (Accuracy 2): The detected BPM is within 4% of the ground truth or any octave-related multiple (2x, 0.5x, 3x, 1/3x).
  - Standard octave-tolerant metric used in MIREX and ISMIR evaluations.
- **Octave errors**: Tracks that pass Acc2 but fail Acc1, indicating the detector found the correct tempo at the wrong metrical level.

6.3 Execution

```
./bench/run_benchmark.sh
```

The benchmark script processes each track through the `open-bpm` release binary and compares the output against ground-truth annotations. Per-track results are written to `bench/benchmark_results.tsv`.

---

7. Context and Comparison

For reference, state-of-the-art deep-learning tempo estimators typically achieve Acc1 scores in the 80–95% range on GiantSteps, while traditional signal-processing methods generally fall in the 60–80% range.

open-bpm's Acc1 of **68.8%** and Acc2 of **78.7%** place it competitively among signal-processing approaches, with the advantage of being a lightweight, dependency-free Rust implementation with no neural network inference required.

Key Strengths

- Zero detection failures across all 664 tracks
- High precision when correct: 61.3% of tracks have < 1% error
- Strong performance in the 120–179 BPM range (core electronic music)
- Pure signal processing, no ML dependencies

Known Limitations

- Slow-tempo octave confusion (< 90 BPM, Acc1 = 4.4%)
- Weak 90–119 BPM range performance (Acc1 = 26.8%)
- Mean absolute error skewed by extreme outliers on slow tracks

---

8. Validation of Candidate Improvement Metrics

Two new candidate metrics were implemented and empirically validated against the baseline to determine whether they could improve the pipeline:

8.1 Methodology

For each of the 664 tracks, the following metrics were computed at four BPM points: ground truth, detected, half of detected, and double of detected. The validation tool (`src/bin/validate_metrics.rs`) outputs a TSV with all values; analysis was done with awk.

A metric is considered useful if: 1. On **PASS tracks** (where the detector is already correct), it correctly defends the detected BPM over the half/double alternatives in >80% of cases 2. On **FAIL tracks**, it correctly identifies the ground truth BPM in >60% of cases

8.2 Harmonic Template Score

Idea: at the true fundamental, ACF harmonics (h1, h2, h3, h4) should decay monotonically and the sub-harmonic (lag/2) should be weaker than the fundamental.

Test	Result
PASS tracks: HT defends det over alternatives	**7%**
PASS tracks: HT incorrectly prefers double	81%
PASS tracks: HT incorrectly prefers half	41%
FAIL tracks: HT prefers GT (correct)	21%
FAIL tracks: HT prefers detected (wrong)	69%

Verdict: The metric is anti-correlated with truth. The formulation clamps `(h2/h1)` at 1.0, which destroys the signal that distinguishes octave errors. Marked as broken in the source; kept as a starting point for future redesign.

8.3 MDL Score

Idea: model the onset sequence as deviations from a regular grid at each BPM candidate. Lower total bits = better fit.

Test	Result
PASS tracks: MDL defends det over alternatives	**29%**
FAIL tracks: MDL prefers GT (correct)	52%

Verdict: Essentially random. The residual magnitudes (~10^5 – 10^6) are not properly normalized, and the data cost dominates the model cost. Marked as needing redesign.

8.4 Cascaded Rejection (Round 2)

Both metrics were integrated into a cascaded rejection scheme: when fusion confidence < 0.55, a weighted composite score (HT * 0.35 + MDL * 0.25 + grid * 0.20 + zone * 0.10 + stability) selected among candidate BPMs.

Result on full GiantSteps benchmark: Acc1 dropped from 68.8% to **32.0%**. The pipeline was reverted to baseline immediately. The validation step (above) was added afterward to identify the root cause.

8.5 Round 3: Four New Deterministic Metrics

After Round 2, four new candidate metrics were designed with the explicit goal of being deterministic, parameter-free, and orthogonal in the signals they capture:

Metric	Mathematical basis	What it measures
Phase coherence R	Rayleigh statistic / circular mean	Angular concentration of onsets at the candidate period
Empty slot score	Combinatorial coverage	Fraction of grid slots filled by at least one onset
Median energy ratio	Robust statistics	Energy separation between on-grid and off-grid onsets
IOI multiple score	Arithmetic divisibility	Fraction of inter-onset intervals fitting integer ratios of T

All four were validated empirically on the same 664 tracks before any integration. Results:

Metric	PASS defense	FAIL recovery	Verdict
Phase coherence R	12.7%	54.6%	Failed
Empty slot score	25.8%	47.8%	Failed
Median energy ratio	14.2%	49.8%	Failed
IOI multiple score	1.1%	38.2%	Failed

None passed the 80% / 60% threshold. None integratable.

8.6 The Structural Wall

A diagnostic analysis revealed the underlying cause. Across all 678 evaluated rows, **phase coherence at the doubled BPM was higher than at the detected BPM in 78% of cases** (and more than 2x higher in 56% of cases). This is not noise – it is a structural property:

At any correct BPM, the same musical content is also periodic at integer multiples of that BPM. A kick drum at 140 BPM is mathematically also "phase-locked" to a 280 BPM grid (every other slot is filled). When evaluated at the doubled BPM:

- The grid spacing is half, so the fixed 30ms tolerance becomes a larger fraction of the period
- Random transients (hi-hat tails, snare crack, ghost notes) have twice as many slots to potentially align with
- The doubled BPM mathematically *contains* the structure of the correct BPM as a subset

This implies a **fundamental limit on onset-only signal processing**: no scalar metric defined as a function of `(onset_times, candidate_period)` alone can reliably distinguish a correct BPM from its integer-multiple harmonics, because the harmonic structure is not observable from raw onset positions.

8.7 Implications

The 68.8% Acc1 baseline appears to represent the practical ceiling of onset-only signal processing on this dataset. Further accuracy gains require information that the onset domain does not contain:

1. **Source separation** – isolate the kick from other instruments before computing periodicities. Current low-band autocorrelation approximates this but adding it to fusion regressed in Round 1.
2. **Higher-level features** – chord change detection, structural boundary analysis, downbeat estimation. These require independent analysis pipelines.
3. **Learned judge router** – train a small classifier on (track features, candidate BPMs, ground truth) tuples from external datasets (Ballroom, GTZAN, Hainsworth) to learn which estimator to trust per track type.

8.8 Lessons Learned

1. **Empirical validation is mandatory before pipeline integration.** Theoretical soundness does not guarantee practical discrimination. Three rounds of attempts confirm this.
2. **Hand-tuned thresholds are an antipattern at this point.** Every parametric tweak (Round 1) and every composite scoring scheme (Round 2) regresses because the system is at a local optimum reachable from this design space.
3. **The structural wall is real.** Round 3 proved by direct measurement that onset-only metrics cannot distinguish a fundamental from its harmonics. Going beyond requires either source separation or external supervision.

---

9. Raw Data

Full per-track results are available in `bench/benchmark_results.tsv` with columns:

Column	Description
`track_id`	GiantSteps track identifier
`ground_truth`	Annotated BPM
`detected`	open-bpm output BPM
`error_pct`	Signed relative error (%)
`acc1`	PASS/FAIL for 4% tolerance
`acc2`	PASS/FAIL for octave-tolerant 4%